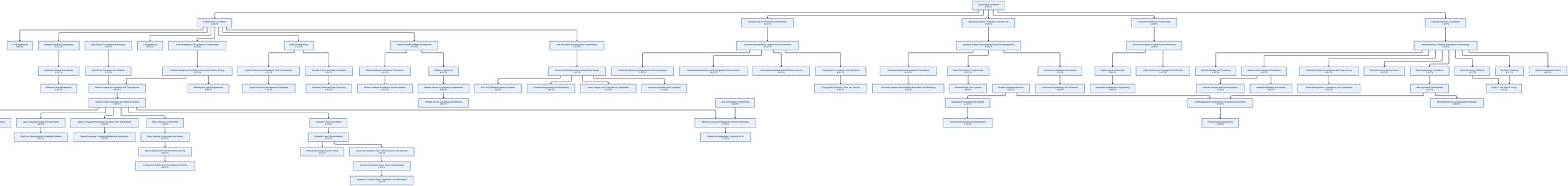

Computing Foundations 6 ECTS
Programming Foundations 6 ECTS
Cybersecurity Fundamentals and Protection 6 ECTS
Operating Systems Foundations and Services 4 ECTS
Computer Architecture Fundamentals 6 ECTS
Computer Networks Foundations 4 ECTS
C++ Programming 6 ECTS
Database Systems Fundamentals 5 ECTS
Data Science Foundations and Analysis 6 ECTS
C Programming 6 ECTS
Artificial Intelligence Foundations: Fundamentals 6 ECTS
Python Programming 6 ECTS
Object-Oriented Software Development 6 ECTS
Data Structures and Algorithms Fundamentals 6 ECTS
Cybersecurity Application, Detection, and Governance 3 ECTS
Operating Systems Processes and Resource Management 5 ECTS
Advanced Processor Execution and Performance 3 ECTS
Internetworking, Transport, and Network Programming 5 ECTS
Database Systems and Internals 4 ECTS
Data Mining Processes and Validation 3 ECTS
Artificial Intelligence Foundations: Advanced Problem Solving 3 ECTS
Digital Productivity and Research Tools Fundamentals 5 ECTS
Scientific Python and Data Foundations 3 ECTS
Modern Software Architecture Foundations 6 ECTS
Software Engineering 6 ECTS
Advanced Data Structures and Algorithmic Design 3 ECTS
Advanced Cybersecurity Approaches and Cryptography 3 ECTS
Federated Authentication and Cryptographic Communication 3 ECTS
Vulnerability Assessment and Offensive Security 3 ECTS
Cryptographic Foundations and Algorithms 4 ECTS
Virtualized Systems Administration Foundations 5 ECTS
Real-Time Systems Fundamentals 2 ECTS
Concurrency Models and Formalisms 2 ECTS
Digital Logic Fundamentals 5 ECTS
Digital Systems and Cryptographic Arithmetic 6 ECTS
Big Data Storage and Processing 6 ECTS
Network Communication Foundations 3 ECTS
Distributed Systems Foundations and Programming 4 ECTS
Blockchain and Cryptocurrencies 6 ECTS
Web Programming Foundations 3 ECTS
Cloud Computing Platforms 4 ECTS
Big Data Processing 5 ECTS
Network Analysis and Testing 6 ECTS
Advanced Data Management 6 ECTS
Machine Learning Foundations and Core Methods 6 ECTS
Planning and Agent Programming 6 ECTS
Digital Productivity and Research Workflows 4 ECTS
Scientific Python and Data Computing 6 ECTS
Modern Software Architecture Service Systems 3 ECTS
Reliable Service Engineering Fundamentals 6 ECTS
SAT and Satisfiability Modulo Theories 3 ECTS
Constraint Programming and Scheduling 3 ECTS
Linear, Integer, and Combinatorial Optimization 3 ECTS
Advanced Algorithms and Complexity 6 ECTS
Cryptographic Protocols, Trust, and Security 5 ECTS
Virtualized Systems Administration Automation and Monitoring 4 ECTS
Advanced Real-Time Systems 3 ECTS
Human-Computer Interaction 6 ECTS
Concurrent Programming and Verification 2 ECTS
Embedded Systems and Programming 4 ECTS
Internet Services and Protocol Analysis 3 ECTS
Wireless Networking and Mobility 3 ECTS
Distributed Algorithms, Consistency, and Coordination 5 ECTS
Web Application Development 6 ECTS
Edge IoT and Web of Things 6 ECTS
Decision Trees, Ensembles, and Model Evaluation 3 ECTS
Reliable Service Governance and Maturity 3 ECTS
Java and Modular Programming 9 ECTS
Geospatial Foundations and Analysis 4 ECTS
Mobile Application Development Foundations and Android 6 ECTS
Web Standards and Cryptographic Arithmetic 6 ECTS
Machine-Learning Platforms and Models 6 ECTS
Large Language Models and Applications 6 ECTS
Natural Language Processing Foundations and Text Analysis 5 ECTS
Deep Learning Fundamentals 4 ECTS
Computer Vision Foundations 6 ECTS
Machine-Learning Workloads and Model Optimization 4 ECTS
Context-Aware Systems and Applications 5 ECTS
iOS Application Development 3 ECTS
Graph and Neurosymbolic Knowledge Systems 3 ECTS
Natural Language Processing Models and Applications 4 ECTS
Deep Learning Architectures and Models 5 ECTS
Computer Vision Scene Analysis 3 ECTS
Parallel and Accelerated Computing for AI 5 ECTS
Agentic Systems and Reinforcement Learning 5 ECTS
Medical Technology and 3D Printing 6 ECTS
Advanced Computer Vision: Geometry and Local Methods 3 ECTS
AI Alignment, Safety, and Computational Creativity 4 ECTS
Advanced Computer Vision: Visual Understanding 3 ECTS
Advanced Computer Vision: Generation and Robustness 3 ECTS